SDN - Topology Detector

Screenshots

Pox Connector

```
ubuntu@vmUbuntu:~/pox$ python3.9 ./pox.py openflow.discovery topology_detector_pox
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
WARNING:version:Support for Python 3 is experimental.
INFO:core:POX 0.7.0 (gar) is up.
INFO:openflow.of_01:[00-00-00-00-00-01 2] connected
INFO:topology_detector_pox:2026-04-20 06:24:34.699598 - Switch ADDED: 1
INFO:openflow.of_01:[00-00-00-00-00-03 3] connected
INFO:topology_detector_pox:2026-04-20 06:24:34.744528 - Switch ADDED: 3
INFO:openflow.of_01:[00-00-00-00-00-02 4] connected
INFO:topology_detector_pox:2026-04-20 06:24:34.746887 - Switch ADDED: 2
INFO:openflow.discovery:link detected: 00-00-00-00-00-03.3 -> 00-00-00-00-00-01.2
INFO:topology_detector_pox:2026-04-20 06:24:34.764592 - Link ADDED: (3, 1)
INFO:openflow.discovery:link detected: 00-00-00-00-00-02.3 -> 00-00-00-00-00-01.1
INFO:topology_detector_pox:2026-04-20 06:24:34.793046 - Link ADDED: (2, 1)
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
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ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
ERROR:packet:(dns) parsing answers: ord() expected string of length 1, but int found
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
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Apr 23 13:29
ubuntu@vmUbuntu: ~/pox
ubuntu@vmUbuntu: ~/pox    ubuntu@vmUbuntu: ~/pox
INFO:topology_detector_pox:2026-04-23 13:27:56.071299 - Link REMOVED: (3, 1)
INFO:topology_detector_pox:2026-04-23 13:27:56.071505 - Link REMOVED: (1, 3)
INFO:topology_detector_pox:2026-04-23 13:27:56.071654 - Link REMOVED: (1, 2)
INFO:topology_detector_pox:2026-04-23 13:27:56.071764 - Link REMOVED: (2, 1)
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
INFO:openflow.discovery:link detected: 00-00-00-00-00-03.3 -> 00-00-00-00-00-01.2
INFO:topology_detector_pox:2026-04-23 13:27:57.996436 - Link ADDED: (3, 1)
INFO:openflow.discovery:link detected: 00-00-00-00-00-01.2 -> 00-00-00-00-00-03.3
INFO:topology_detector_pox:2026-04-23 13:27:58.639346 - Link ADDED: (1, 3)
INFO:openflow.discovery:link detected: 00-00-00-00-00-01.1 -> 00-00-00-00-00-02.3
INFO:topology_detector_pox:2026-04-23 13:27:59.714468 - Link ADDED: (1, 2)
INFO:openflow.discovery:link detected: 00-00-00-00-00-02.3 -> 00-00-00-00-00-01.1
INFO:topology_detector_pox:2026-04-23 13:28:00.729530 - Link ADDED: (2, 1)
INFO:openflow.discovery:link timeout: 00-00-00-00-00-03.3 -> 00-00-00-00-00-01.2
INFO:openflow.discovery:link timeout: 00-00-00-00-00-01.2 -> 00-00-00-00-00-03.3
INFO:openflow.discovery:link timeout: 00-00-00-00-00-01.1 -> 00-00-00-00-00-02.3
INFO:openflow.discovery:link timeout: 00-00-00-00-00-02.3 -> 00-00-00-00-00-01.1
INFO:topology_detector_pox:2026-04-23 13:28:14.975008 - Link REMOVED: (3, 1)
INFO:topology_detector_pox:2026-04-23 13:28:14.975318 - Link REMOVED: (1, 3)
INFO:topology_detector_pox:2026-04-23 13:28:14.975579 - Link REMOVED: (1, 2)
INFO:topology_detector_pox:2026-04-23 13:28:14.975821 - Link REMOVED: (2, 1)
INFO:openflow.discovery:link detected: 00-00-00-00-00-03.3 -> 00-00-00-00-00-01.2
INFO:topology_detector_pox:2026-04-23 13:28:16.237941 - Link ADDED: (3, 1)
INFO:openflow.discovery:link detected: 00-00-00-00-00-01.2 -> 00-00-00-00-00-03.3
INFO:topology_detector_pox:2026-04-23 13:28:16.862100 - Link ADDED: (1, 3)
INFO:openflow.discovery:link detected: 00-00-00-00-00-01.1 -> 00-00-00-00-00-02.3
INFO:topology_detector_pox:2026-04-23 13:28:17.491417 - Link ADDED: (1, 2)
INFO:openflow.discovery:link detected: 00-00-00-00-00-02.3 -> 00-00-00-00-00-01.1
```
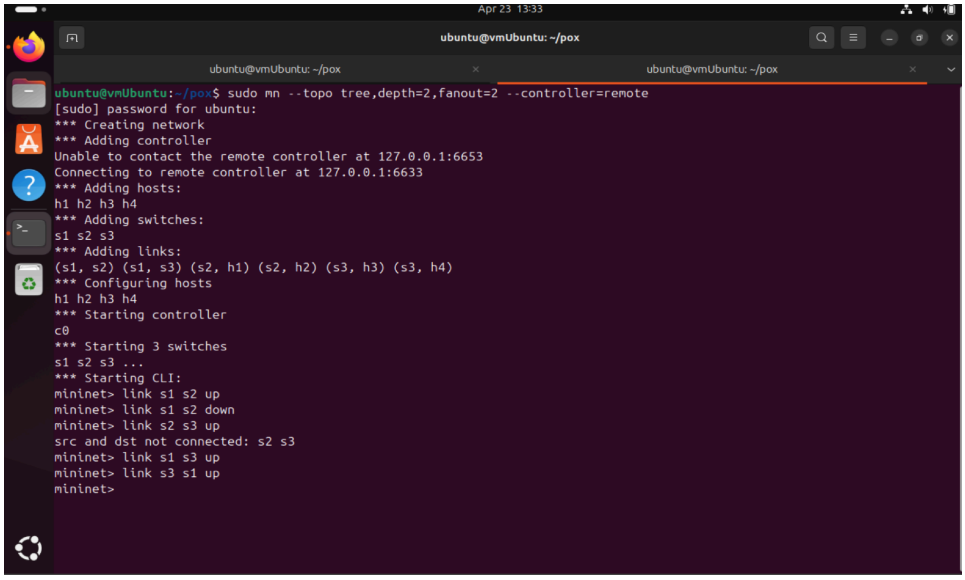

Mininet


Apr 23 13:33
ubuntu@vmUbuntu: ~/pox
ubuntu@vmUbuntu: ~/pox
ubuntu@vmUbuntu: ~/pox
ubuntu@vmUbuntu:~/pox$ sudo mn --topo tree,depth=2,fanout=2 --controller=remote
[sudo] password for ubuntu:
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
Connecting to remote controller at 127.0.0.1:6633
*** Adding hosts:
h1 h2 h3 h4
*** Adding switches:
s1 s2 s3
*** Adding links:
(s1, s2) (s1, s3) (s2, h1) (s2, h2) (s3, h3) (s3, h4)
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller
c0
*** Starting 3 switches
s1 s2 s3 ...
*** Starting CLI:
mininet> link s1 s2 up
mininet> link s1 s2 down
mininet> link s2 s3 up
src and dst not connected: s2 s3
mininet> link s1 s3 up
mininet> link s3 s1 up
mininet>